



Understanding Drivers and Barriers to Electric Vehicle Adoption in Indonesia Through Social Media Mining: A Policy-Oriented Analysis

Dwi Adi Purnama^{1,2*}

¹Department of Industrial Engineering, Faculty of Industrial Technology, Universitas Islam Indonesia, 55584 Yogyakarta, Indonesia

²Department of Engineering Management, Faculty of Industrial Technology, Universitas Islam Indonesia, 55584 Yogyakarta, Indonesia

* Correspondence: Dwi Adi Purnama (dwiadipurnama@uii.ac.id)

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Abstract: The global transition toward electric vehicles has been recognized as a critical pathway for achieving sustainable transportation and reducing greenhouse gas emissions. In Indonesia, a series of policy interventions has been introduced to accelerate market adoption. Nevertheless, the effectiveness of these initiatives has been constrained by public acceptance, which remains a decisive factor in the diffusion of electric vehicle technologies. Although public perceptions have traditionally been examined through surveys, the dynamic and large-scale opinions expressed on social media remain underexplored in the Indonesian context. To address this gap, public discourse surrounding electric vehicle adoption in Indonesia was investigated through a social media mining framework. A total of 6,844 publicly available posts published on X (formerly Twitter) over a one-year period were collected using five electric vehicle-related keywords, namely electric vehicles, electric cars, electric motorcycles, electric buses, and electric vehicle batteries. Public sentiment was classified using the IndoRoBERTa sentiment analysis model, while latent discussion themes were identified through topic modeling to characterize the principal drivers and barriers influencing electric vehicle adoption. The results indicate that positive public perceptions were primarily associated with environmental sustainability, lower operating costs, continued expansion of charging infrastructure, and government incentives. In contrast, negative perceptions were predominantly linked to inadequate charging infrastructure, inconsistencies in policy implementation, concerns regarding technological reliability, uncertainties associated with the transition from conventional vehicles, and unfavorable user experiences. Building upon these findings, a policy-oriented analytical framework was established to systematically translate large-scale social media discourse into evidence-based policy priorities for accelerating electric vehicle adoption in Indonesia. The findings demonstrate that social media analytics can provide timely and complementary insights into evolving public perceptions that are difficult to capture through conventional survey-based approaches. An interpretable decision-support framework is provided for policymakers to identify emerging public concerns, evaluate policy effectiveness, and design adaptive strategies that promote sustainable transportation and facilitate the long-term transition toward electric mobility in Indonesia.

Keywords: Electric vehicle adoption; Social media mining; Sustainable transportation; Public perception; Transport policy; Indonesia.

1. Introduction

The transportation sector plays a critical role in economic development by facilitating the movement of people and goods. However, it is also one of the largest contributors to greenhouse gas emissions, energy consumption, and urban air pollution worldwide. According to the International Energy Agency, transportation accounts for nearly one-quarter of global energy-related carbon dioxide emissions, making the decarbonization of transport

systems a central component of sustainable development strategies (IEA, 2024). The growing urgency to mitigate climate change and improve urban environmental quality has accelerated the transition toward low-carbon mobility solutions, among which electric vehicles have emerged as one of the most promising alternatives.

Electric vehicles are widely recognized for their potential to reduce greenhouse gas emissions, decrease dependence on fossil fuels, improve energy efficiency, and support sustainable transportation systems. Consequently, governments across the world have implemented various policy measures to accelerate electric vehicle adoption, including financial incentives, tax reductions, charging infrastructure development, and industrial investment programs. Previous studies have shown that electric vehicle deployment can significantly contribute to emission reduction targets while simultaneously supporting energy security and industrial competitiveness (Asamer et al., 2016; Degirmenci & Breitner, 2017; Rezvani et al., 2015).

In Southeast Asia, the transition toward electric mobility has gained increasing attention as governments seek to balance economic growth with environmental sustainability. Indonesia represents a particularly important case due to its large population, rapidly expanding middle class, growing automotive market, and ambitious commitment to becoming a regional electric vehicle manufacturing hub (Anugerah et al., 2021; Purnama et al., 2025). Through Presidential Regulation No. 55/2019 and subsequent policy initiatives, the Indonesian government has actively promoted the development of electric vehicle ecosystems, including battery manufacturing, charging infrastructure expansion, fiscal incentives, and industrial investment support. These initiatives are expected to accelerate electric vehicle adoption while contributing to national decarbonization goals and sustainable transportation development.

Despite these efforts, electric vehicle adoption in Indonesia remains at an early stage compared with more mature markets such as China, Norway, and several European countries. Previous studies indicate that electric vehicle adoption is influenced by multiple factors, including environmental awareness, perceived economic benefits, charging infrastructure availability, technological reliability, government incentives, and social influence (Balla et al., 2023; Hardman et al., 2017; Nugroho & Widiyanto, 2024). At the same time, concerns regarding vehicle affordability, charging accessibility, battery durability, maintenance costs, and policy uncertainty continue to hinder broader market acceptance (Purnama et al., 2025). These findings suggest that electric vehicle adoption is not merely a technological transition but a complex socio-technical process involving interactions among consumers, policymakers, manufacturers, infrastructure providers, and energy systems. Understanding public perceptions is therefore essential for designing effective transportation policies that support electric vehicle adoption. Public perceptions influence consumer behavior, technology acceptance, investment decisions, and policy legitimacy (Hardman et al., 2017; Li et al., 2017; Rezvani et al., 2015). Most existing studies investigating electric vehicle adoption in Indonesia have relied primarily on survey-based approaches using frameworks such as the technology acceptance model (Davis, 1989), unified theory of acceptance and use of technology (Venkatesh et al., 2003), and diffusion of innovation theory. While these studies provide valuable insights into consumer intentions and behavioral determinants, they often face limitations related to sample size, geographical coverage, self-reporting bias, and their limited ability to capture dynamic public discourse in real time (Acheampong & Cugurullo, 2019; Balla et al., 2023; Stieglitz et al., 2018).

The rapid growth of social media platforms offers new opportunities for understanding public perceptions at an unprecedented scale. Platforms such as X (formerly Twitter) have become important channels through which individuals express opinions, share experiences, discuss policy issues, and evaluate emerging technologies. Social media mining enables researchers to collect and analyze large volumes of user-generated content, providing timely and naturally occurring insights into public attitudes. In transportation research, social media analytics has been increasingly used to investigate public perceptions of shared mobility, autonomous vehicles, public transportation systems, and electric vehicles (Acheampong & Cugurullo, 2019; Balla et al., 2023; Stieglitz et al., 2018).

Among various social media analytics techniques, sentiment analysis and topic modeling have proven particularly useful for extracting meaningful information from large textual datasets. Sentiment analysis allows researchers to evaluate public attitudes toward specific technologies, while topic modeling identifies dominant themes and concerns underlying online discussions (Subagyo et al., 2024). Recent studies have demonstrated that combining sentiment analysis with topic modeling can provide deeper insights into consumer perceptions, policy acceptance, and technology adoption barriers (Choi et al., 2020; Jeong et al., 2019; Wang et al., 2023). Nevertheless, existing electric vehicle-related social media studies predominantly focus on either public sentiment or policy evaluation separately. Limited research has systematically identified both adoption drivers and barriers while translating these findings into actionable transportation policy recommendations, particularly in emerging electric vehicle markets such as Indonesia. This research gap is important because effective electric vehicle policy requires not only understanding whether public sentiment is positive or negative, but also identifying the specific factors that encourage or hinder adoption. Policymakers require evidence-based insights regarding infrastructure needs, financial incentives, technological concerns, environmental motivations, and user experiences to design targeted interventions that accelerate electric vehicle diffusion. Without such understanding, policy initiatives may fail to address the most critical barriers perceived by potential users.

To address this gap, this study investigates public perceptions of electric vehicle adoption in Indonesia through social media mining. Based on these gaps, a policy-oriented analytical framework is developed to translate public discourse into policy priorities for accelerating electric vehicle adoption in Indonesia. The study contributes to the literature in three ways. First, it provides large-scale empirical evidence regarding public perceptions of electric vehicles in an emerging market context. Second, it integrates sentiment analysis and topic modeling to simultaneously identify adoption drivers and barriers from social media discourse. Third, it develops a policy-oriented perspective that transforms online public opinions into actionable recommendations for sustainable transportation planning and electric mobility development. Through these contributions, the study seeks to support policymakers, transportation planners, and industry stakeholders in designing more effective strategies for accelerating the transition toward sustainable mobility in Indonesia.

2. Literature Reviews

2.1 Electric Vehicle Adoption and Sustainable Transportation

The transition toward electric vehicles has become a central component of sustainable transportation strategies worldwide. Electric vehicles are widely regarded as an effective solution for reducing greenhouse gas emissions, improving energy efficiency, decreasing dependence on fossil fuels, and supporting climate mitigation targets. Consequently, governments across developed and developing countries have introduced various policy measures to accelerate electric vehicle adoption, including purchase subsidies, tax incentives, charging infrastructure development, and industrial investment programs (Degirmenci & Breitner, 2017; Hardman et al., 2017; Rezvani et al., 2015).

Previous studies have identified electric vehicle adoption as a complex socio-technical transition involving interactions among consumers, technology providers, policymakers, energy systems, and transportation infrastructure. Unlike conventional vehicle adoption, electric vehicle adoption depends not only on product attributes but also on ecosystem readiness, including charging accessibility, electricity supply, regulatory support, and public confidence in emerging technologies (Li et al., 2017). As a result, understanding public acceptance has become a critical prerequisite for effective transportation policy formulation.

Recent studies have further strengthened the understanding of electric vehicle adoption in emerging markets, particularly Indonesia. Astuti & Susanto (2024) showed that electric vehicle purchase intention in Indonesia is shaped not only by perceived usefulness and ease of use, but also by perceived risk, electric vehicle knowledge, and public involvement, indicating that adoption is driven by both cognitive and contextual factors. Hakam & Jumayla (2024) similarly argued that Indonesia's electric vehicle transition remains constrained by infrastructure readiness, policy consistency, and affordability, despite strong long-term potential. In a more focused barrier analysis, Lazuardy et al. (2024) identified technological, environmental, economic, and regulatory barriers as the main obstacles to electric vehicle diffusion in Indonesia. Complementing these studies, performance expectancy, environmental concern, charging infrastructure, and financial incentives positively influence consumer attitudes, while price and operating costs remain significant barriers. Collectively, these studies confirm that electric vehicle adoption in Indonesia is a multidimensional issue involving technology perceptions, infrastructure availability, cost concerns, and institutional support.

2.2 Drivers and Barriers of Electric Vehicle Adoption

The literature identifies numerous factors influencing electric vehicle adoption. Environmental benefits are among the most frequently reported drivers. Consumers often perceive electric vehicles as environmentally friendly alternatives that contribute to reducing air pollution and carbon emissions (Rezvani et al., 2015). Economic benefits, including lower fuel costs and reduced maintenance expenses, also play important roles in encouraging adoption (Hardman et al., 2017). Infrastructure development has emerged as another critical determinant. The availability of public charging stations significantly reduces range anxiety and increases consumer confidence in electric vehicle usage (Sierzechula et al., 2014). Government support through subsidies, tax reductions, and regulatory incentives has similarly been shown to stimulate electric vehicle adoption by reducing financial barriers and improving perceived attractiveness (Degirmenci & Breitner, 2017).

Despite these positive drivers, several barriers continue to hinder widespread adoption. Previous studies consistently report concerns regarding charging infrastructure availability, battery durability, vehicle affordability, charging time, and technological reliability (Li et al., 2017). In emerging economies, additional challenges include limited public awareness, uncertainty regarding maintenance services, and inconsistent policy implementation. These barriers indicate that electric vehicle adoption remains strongly influenced by both technological and institutional factors. Recent studies conducted in Southeast Asia and Indonesia reveal similar findings. Infrastructure readiness, purchase costs, policy uncertainty, and technological skepticism continue to represent significant obstacles to adoption. However, most existing studies rely on structured survey instruments, limiting

their ability to capture spontaneous public concerns and rapidly changing perceptions (Li et al., 2017). Consequently, additional research is needed to explore public discourse using alternative data sources capable of reflecting real-time societal perspectives using social media data.

2.3 Social Media Mining in Transportation Research

The emergence of social media platforms has transformed the way public opinions can be collected and analyzed. Social media platforms such as X, Facebook, YouTube, and online discussion forums generate vast amounts of user-generated content that reflect consumer experiences, preferences, concerns, and evaluations. These digital traces provide valuable opportunities for researchers to understand public perceptions at a scale that is difficult to achieve through conventional survey methods (Stieglitz et al., 2018). In transportation research, social media mining has increasingly been used to investigate public attitudes toward transportation systems, autonomous vehicles, shared mobility services, public transportation, and emerging transportation technologies (Acheampong & Cugurullo, 2019). Compared with survey-based approaches, social media analytics enables continuous monitoring of public discourse, broader geographical coverage, and the identification of emerging issues in near real time.

Recent international studies highlight the growing importance of public discourse, spatial variation, and policy signals in shaping electric vehicle adoption. Qian & Gkritza (2024) demonstrated that public perceptions of electric vehicles vary across regions and over time, suggesting that electric vehicle strategies should be tailored to local adoption contexts. Wang et al. (2024) further showed that public sentiment toward electric vehicles and charging infrastructure is influenced by geographic conditions, charging station availability, and infrastructure equity, especially in rural areas. Zhao et al. (2020) revealed that media attention can significantly accelerate electric vehicle adoption, suggesting that public information environments also play an important role in market diffusion. Ashby et al. (2025) added that public views on electric vehicle charging are strongly shaped by charger reliability, accessibility, blocked chargers, climate-related performance, and charging inequality. At a broader level, Taamneh & Makahleh (2025) synthesized the urban electric vehicle adoption literature and confirmed that infrastructure scarcity, range anxiety, and charging time continue to be persistent barriers in many cities. Similarly, Alshahapy et al. (2025) emphasized that current electric vehicle adoption challenges are no longer limited to consumer acceptance alone, but also involve supply-chain constraints, infrastructure inequity, and broader transition justice issues.

Nevertheless, challenges remain regarding data quality, representativeness, and interpretation. Social media users may not fully represent the broader population, while textual data often require analytical techniques to extract meaningful insights. Despite these limitations, social media mining continues to gain importance as a scalable approach for understanding public perceptions and supporting evidence-based transportation policies (Balla et al., 2023; Purnama et al., 2025).

2.4 Sentiment Analysis, Topic Modeling, and Policy-Oriented Social Media Analytics

Among various social media analytics techniques, sentiment analysis and topic modeling are the most widely applied methods for extracting information from large textual datasets. Sentiment analysis enables researchers to identify public attitudes toward products, services, technologies, or policies, while topic modeling reveals latent themes embedded within online discussions (Stieglitz et al., 2018). Jeong et al. (2019) demonstrated how social media mining can be used to identify customer needs and product opportunities through the integration of sentiment analysis and topic modeling. Similarly, Choi et al. (2020) showed that online discourse can reveal evolving customer preferences and emerging opportunities for innovation. These studies highlight the potential of social media analytics as a mechanism for transforming large-scale public discourse into actionable managerial insights.

Recent transportation studies have increasingly adopted these approaches to analyze perceptions of electric vehicles. However, most existing studies focus primarily on sentiment classification, adoption intention, or policy evaluation separately. Few studies have systematically integrated sentiment analysis and topic modeling to identify both drivers and barriers of electric vehicle adoption while simultaneously translating these findings into transportation policy recommendations. This limitation is particularly important in emerging electric vehicle markets such as Indonesia, where policymakers require evidence-based insights regarding infrastructure development, financial incentives, technological concerns, environmental motivations, and user experiences (Purnama et al., 2025). Understanding these issues is essential for designing targeted interventions capable of accelerating electric vehicle adoption and supporting sustainable transportation transitions.

2.5 Research Gap

Based on the reviewed literature, several research gaps can be identified (Table 1). First, most electric vehicle adoption studies continue to rely on survey-based methods, limiting their ability to capture large-scale and dynamic public discourse. Second, existing social media studies often focus on either sentiment analysis or topic identification without comprehensively examining both adoption drivers and adoption barriers. Third, limited research has translated social media findings into actionable transportation policy priorities. Accordingly, this study addresses these gaps by integrating sentiment analysis and topic modeling to investigate public perceptions of electric vehicle adoption in Indonesia and develop policy-oriented recommendations for accelerating the electric mobility transition.

Table 1. State-of-the-art classification of electric vehicle adoption and social-media-based policy studies

Reference	Social Media Data	Survey Data	Topic Modeling	Sentiment Analysis	Theory Used	Drivers	Barriers	Policy Analysis	Research Object
Rezvani et al. (2015)					Review of electric vehicle adoption literature	√	√		Consumer electric vehicle adoption research
Jeong et al. (2019)	√		√	√	Not explicit				Product opportunity mining from social media
Astuti & Susanto (2024)		√			Extended technology acceptance model/purchase intention model	√	√		Electric vehicle purchase intention in Indonesia
Hakam & Jumayla (2024)					Review/comparative policy perspective	√	√	√	Electric vehicle adoption in Indonesia
Lazuardy et al. (2024)		√			Barrier framework		√	√	Electric vehicle adoption barriers in Indonesia
Nugroho & Widiyanto (2024)	√			√	Zero-shot aspect-based sentiment analysis	√	√		Electric vehicle adoption in Indonesia using Twitter data
Qian & Gkritza (2024)	√		√	√	Not explicit	√	√		Public perception of electric vehicles using Twitter data
Wang et al. (2024)	√			√	Not explicit		√		Electric vehicle and charging infrastructure sentiment from Twitter data
Bachtiar et al. (2024)		√			Partial least squares structural equation modeling/adoption intention model	√	√	√	Market adoption of electric vehicles in Indonesia

Ashby et al. (2025)	√	√	√	Thematic analysis	√	√	Public views on electric vehicle charging Drivers of electric vehicle adoption for sustainable development
Bansal et al. (2024)	√			Structural equation modeling–artificial neural network/sustainable development drivers	√		Electric vehicle adoption effectiveness in Indonesia
Basmantra et al. (2025)	√			Protection motivation theory + green branding and policy framework	√	√	Public perceptions of electric vehicle adoption in Indonesia through social media mining
This study	√	√	√	Policy-oriented social media mining framework	√	√	

3. Methodology

This study adopts a policy-oriented social media mining approach to investigate the drivers and barriers of electric vehicle adoption in Indonesia. The methodological framework is designed to transform large-scale online public discourse into evidence-based policy insights. In line with recent studies on digital public opinion and text analytics, the framework consists of five main phases: (i) online data collection and screening, (ii) data preprocessing, (iii) sentiment analysis, (iv) topic modeling and driver–barrier identification, and (v) policy-oriented analysis and evaluation. The overall framework is intended to capture naturally occurring public perceptions regarding electric vehicle adoption and translate them into actionable policy priorities.

3.1 Phase 1: Online Data Collection and Screening

3.1.1 Social media data collection

The study uses social media data from X because the platform provides large volumes of user-generated content reflecting public opinions, experiences, expectations, and concerns regarding transportation technologies. Social media data are particularly useful for electric vehicle research because they allow the observation of spontaneous public discourse outside structured survey settings. A total of 6,844 public posts related to electric vehicles were collected from X during one-year period. Data scraping was conducted using Google Colaboratory and Python-based scraping tools. To ensure coverage of major electric vehicle-related discussions, five keywords were used during data collection: electric vehicles, electric cars, electric motorcycles, electric buses, and electric vehicle batteries. These keywords were selected because they represent major categories of electric vehicle discourse in Indonesia, including private mobility, public transport, general electric vehicle terminology, and battery-related issues.

For reproducibility, the data-processing workflow was implemented in Python in Google Colaboratory. The main libraries used in the workflow included pandas and NumPy for data handling, regular expression functions for text cleaning, scikit-learn for feature preparation and evaluation support, Hugging Face Transformers and PyTorch for the IndoRoBERTa sentiment classifier, and Gensim for Latent Dirichlet Allocation (LDA) topic modeling. The random seed was fixed during model construction to support reproducible topic generation. All posts were stored in a structured dataset containing the original text, cleaned text, keyword category, sentiment label, and topic assignment.

3.1.2 Inclusion and exclusion criteria

To ensure relevance and consistency, inclusion and exclusion criteria were applied during dataset construction. Posts were included if they: (i) contained at least one of the selected electric vehicle-related keywords, (ii) were posted within the study period, and (iii) contained textual content relevant to electric vehicle products, services, adoption, infrastructure, public attitudes, or related policy issues. Posts were excluded if they: (i) were duplicates, (ii) contained only links or non-informative text, (iii) were dominated by spam-like promotional content, or (iv) were irrelevant to the context of electric vehicle adoption in Indonesia. Because the study focuses on public

discourse, the final dataset was intended to reflect authentic online opinions rather than repetitive or non-substantive content. The resulting corpus was then used as the basis for sentiment classification and topic extraction.

3.2 Phase 2: Data Preprocessing

Text preprocessing was conducted to improve data quality and prepare the corpus for machine learning analysis. Social media data are typically noisy and unstructured, containing uniform resource locators, hashtags, user mentions, emojis, repeated characters, and inconsistent writing styles. Therefore, the raw dataset underwent several preprocessing stages. First, all text was converted into lowercase to ensure lexical consistency. Second, uniform resource locators, user mentions, hashtags, punctuation, numbers, and special characters were removed. Third, duplicate posts were eliminated to avoid repetitive bias in the dataset. Fourth, text normalization was conducted to reduce lexical variation, including the correction of repeated characters and common non-standard expressions. Fifth, tokenization was applied to segment the text into individual tokens. Finally, stopwords that did not contribute meaningful semantic information were removed. This preprocessing stage was necessary to reduce noise, improve classification performance, and generate more coherent topic structures. The cleaned dataset was then used for sentiment analysis and topic modeling.

3.3 Phase 3: Sentiment Analysis

3.3.1 Sentiment classification

Sentiment analysis was used to classify public opinions regarding electric vehicle adoption into three polarity categories: positive, neutral, and negative. This stage aims to provide an overall view of how electric vehicle-related issues are perceived by the public and to separate supportive discourse from critical discourse. This study employs a fine-tuned IndoRoBERTa-based sentiment classifier because transformer-based language models are able to capture contextual meaning more effectively than conventional lexicon-based methods. The model was implemented using the Hugging Face Transformers framework with a PyTorch backend. Each cleaned post was tokenized using the IndoRoBERTa tokenizer with padding and truncation. The maximum sequence length was set to 128 tokens, which was considered sufficient for short social media posts. Sentiment inference was conducted in batches, and the final class label was assigned using the highest softmax probability among positive, neutral, and negative classes.

The sentiment output for each post consisted of the predicted label and class probabilities. Posts with the highest probability for the positive class were interpreted as supportive discourse, while posts with the highest probability for the negative class were interpreted as critical discourse. Neutral posts were retained to describe the overall structure of the conversation, but the driver-barrier analysis focused mainly on positive and negative subsets because these subsets contain clearer evaluative orientations. This procedure was used to reduce ambiguity before the topic-modeling stage and to ensure that the subsequent LDA analysis was aligned with the sentiment orientation of the corpus.

For each post d_i , the IndoRoBERTa encoder transforms the textual input into a contextual representation vector h_i , defined as:

$$h_i = f_{\theta}(d_i)$$

where, $f_{\theta}(\cdot)$ denotes the IndoRoBERTa encoder parameterized by θ , and $h_i \in R^m$ is the contextual embedding of post i .

The embedding vector was then passed through a linear classification layer to obtain the logit score for each sentiment class. For sentiment class c , the logit is given by:

$$z_{ic} = w_c^T h_i + b_c$$

where, $w_c \in R^m$ is the weight vector associated with class c ; $b_c \in R$ is the corresponding bias term; and z_{ic} is the logit score of post i for class c .

The posterior probability that post d_i belongs to sentiment class c was then computed using the softmax function:

$$p_{ic} = \frac{\exp(z_{ic})}{\sum_{k=1}^3 \exp(z_{ik})}, c \in \{1, 2, 3\}$$

where, $c = 1$, $c = 2$, and $c = 3$ correspond to positive, neutral, and negative sentiment, respectively.

The final predicted sentiment label for each post was determined by selecting the class with the highest posterior probability:

$$\hat{y}_i = \arg \max_{c \in \{1,2,3\}} p_{ic}$$

During the fine-tuning process, the model parameters were optimized by minimizing the cross-entropy loss function:

$$L(\theta, W, b) = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^3 y_{ic} \log(p_{ic})$$

where, N is the total number of training instances, y_{ic} is the ground-truth indicator variable that takes the value 1 if post i belongs to class c , and 0 otherwise, while p_{ic} denotes the predicted probability for class c .

Based on this classification procedure, each cleaned post was assigned to one of the three sentiment categories. Posts classified as positive were interpreted as reflecting favorable attitudes, opportunities, or supportive perceptions toward electric vehicle adoption. Posts classified as negative were interpreted as reflecting dissatisfaction, barriers, criticism, or perceived risks. Neutral posts were retained for descriptive purposes but were not emphasized in the driver–barrier interpretation because they generally represent informational or non-evaluative content.

To summarize the sentiment distribution across the dataset, the number of posts classified into sentiment class c was calculated as:

$$N_c = \sum_{i=1}^N 1(\hat{y}_i = c)$$

where, N_c denotes the number of posts assigned to sentiment class c , and $1(\cdot)$ is the indicator function.

The proportion of each sentiment class was then computed using:

$$S_c = \frac{N_c}{N} \times 100$$

where, S_c represents the percentage of posts belonging to class c .

These sentiment proportions provide a descriptive overview of public attitudes toward electric vehicle adoption and serve as the basis for subsequent topic modeling and policy-oriented analysis.

3.3.2 Sentiment distribution analysis

After inference, each cleaned post was assigned one of three sentiment labels. Positive posts were interpreted as reflecting favorable attitudes, opportunities, or supportive perceptions toward electric vehicle adoption. Negative posts were interpreted as reflecting dissatisfaction, barriers, criticism, or perceived risks. Neutral posts were retained for descriptive analysis but were not emphasized in driver-barrier interpretation because they generally represent informational or non-evaluative content. The distribution of positive, neutral, and negative posts was then calculated for the whole dataset and for each electric vehicle keyword category. This step was used to identify whether specific electric vehicle terms were associated with stronger supportive or critical responses.

After classification, the frequency of positive, neutral, and negative posts was calculated for the entire dataset and for each keyword category. This step provides an overview of sentiment distribution across electric vehicle-related topics. The sentiment results were then used as the basis for separating the corpus into positive and negative subsets for the next stage of topic modeling.

3.4 Phase 4: Topic Modeling and Driver–Barrier Identification

3.4.1 Topic modeling using latent Dirichlet allocation

To identify dominant themes in public discourse, this study applies LDA topic modeling. LDA is a probabilistic machine learning method used to uncover latent thematic structures in textual data by identifying word co-occurrence patterns across documents. The method assumes that each document contains a mixture of topics and that each topic is characterized by a probability distribution over words. In LDA, each document is assumed to be

a mixture of latent topics, and each topic is characterized by a distribution over words. The joint probability of the topic distribution θ , topic assignments z , and observed words w , given the hyperparameters α and β , is defined as:

$$p(\theta, z, w \mid \alpha, \beta) = p(\theta \mid \alpha) \prod_{n=1}^N p(z_n \mid \theta) p(w_n \mid z_n, \beta)$$

where, θ represents the topic proportion of a document; z_n is the hidden topic indicator for word w_n ; and w_n is the observed word token. The hyperparameter α controls the sparsity of document–topic mixtures, whereas β controls the sparsity of topic–word distributions.

In this study, topic modeling was conducted separately for positive and negative corpora. This separation was intended to produce a clearer interpretation of electric vehicle adoption drivers and barriers. Positive-topic modeling helps identify enabling factors and favorable perceptions, while negative-topic modeling highlights adoption barriers, dissatisfaction, and critical public concerns. The LDA model was implemented using the Gensim library. The cleaned corpus was converted into a dictionary and bag-of-words representation. Very rare and overly common tokens were removed before model estimation to improve topic interpretability. In the implementation, token filtering used minimum document frequency and maximum corpus-proportion thresholds, while LDA estimation used a fixed random state, multiple passes, and repeated iterations to support stable convergence. The main settings were alpha = “auto”, eta = “auto”, passes = 20, iterations = 400, chunksize = 100, and random state = 42. The top representative words and documents for each topic were then reviewed to assign substantive labels.

3.4.2 Determining the number of topics

To determine the optimal number of topics, several candidate LDA models were estimated and compared using topic coherence and interpretability. Candidate topic numbers were tested separately for the positive and negative corpora. The final topic number was selected by considering three criteria: (i) the coherence score, (ii) the absence of highly overlapping or redundant topics, and (iii) the substantive interpretability of top words and representative posts. Based on these criteria, ten topics were retained for the positive corpus, and ten topics were retained for the negative corpus. Topic coherence measures the semantic interpretability of the top words within a topic. For a topic with a set of top representative words, the coherence score can be defined as:

$$C(t; V^{(t)}) = \sum_{m=2}^M \sum_{l=1}^{m-1} \log \left(\frac{D(v_m^{(t)}, v_l^{(t)}) + 1}{D(v_l^{(t)})} \right)$$

where, $D(v_l^{(t)})$ denotes the number of documents containing word $v_l^{(t)}$; and $D(v_m^{(t)}, v_l^{(t)})$ denotes the number of documents in which both words $v_m^{(t)}$ and $v_l^{(t)}$ co-occur.

A higher coherence value indicates that the topic is more semantically consistent and interpretable. The coherence score was not used mechanically. A topic solution was accepted only when the topics were both statistically coherent and meaningful for electric vehicle adoption analysis. This additional interpretive criterion was important because a model with a marginally higher coherence score may still produce topics that are difficult to translate into policy-relevant drivers and barriers.

3.4.3 Driver and barrier classification

The extracted topics were classified into adoption drivers and adoption barriers by integrating topic prevalence, sentiment orientation, and qualitative validation of representative posts. This procedure was intended to ensure that topic interpretation was not based solely on word co-occurrence patterns, but also on the contextual meaning of discourse. Let D denote the total number of documents and let θ_{dk} represent the probability of topic k in document d , obtained from the LDA model. The overall importance of topic k across the corpus is defined as:

$$WI_k = \sum_{d=1}^D \theta_{dk}$$

where, WI_k is the topic weight or prevalence of topic k . A larger WI_k indicates that the topic appears more prominently in the corpus.

To determine whether a topic acts as a driver or a barrier, each document d is associated with a sentiment label $s_d \in \{+1, 0, -1\}$, where +1, 0, and -1 denote positive, neutral, and negative sentiment, respectively. Based on this sentiment assignment, the positive and negative contributions of topic k are defined as:

$$P_k = \sum_{d=1}^D \theta_{dk} 1(s_d=+1)$$

$$N_k = \sum_{d=1}^D \theta_{dk} 1(s_d=-1)$$

where, $1(\cdot)$ is the indicator function. Thus, P_k measures the extent to which topic k is associated with positive discourse, while N_k measures the extent to which it is associated with negative discourse.

The net orientation of topic k is then calculated as:

$$O_k = \frac{P_k - N_k}{P_k + N_k}$$

where, $O_k \in [-1, 1]$. If $O_k > 0$, topic k is more strongly associated with positive discourse and is therefore interpreted as an adoption driver. If $O_k < 0$, topic k is more strongly associated with negative discourse and is interpreted as an adoption barrier. If $O_k = 0$, the topic is considered neutral or mixed.

Accordingly, the initial classification rule is defined as:

$$C_k = \begin{cases} \text{Driver,} & O_k > 0 \\ \text{Barrier,} & O_k < 0 \\ \text{Mixed,} & O_k = 0 \end{cases}$$

To strengthen interpretive validity, the quantitative classification was complemented by qualitative examination of representative posts. Let R_k denote the set of representative documents with the highest topic membership values for topic k . A validation score for topic k is defined as:

$$V_k = \frac{1}{|R_k|} \sum_{d \in R_k} v_d$$

where, $v_d = 1$ if representative post d is semantically consistent with the assigned topic interpretation, and $v_d = 0$ otherwise. Thus, V_k represents the proportion of representative posts that support the semantic interpretation of topic k .

The final topic classification is therefore determined by combining sentiment orientation and qualitative validation, such that topics with dominant positive orientation and semantically consistent representative posts are classified as adoption drivers, while topics with dominant negative orientation and semantically consistent representative posts are classified as adoption barriers. Under this formulation, topics related to environmental benefits, lower operating costs, technological attractiveness, infrastructure progress, and policy support can be classified as adoption drivers when they exhibit $O_k > 0$. Conversely, topics related to charging infrastructure concerns, affordability, battery durability, maintenance problems, policy inconsistency, and negative user experiences can be classified as adoption barriers when they exhibit $O_k < 0$.

This formulation enables the study to combine three complementary layers of evidence, namely topic salience from LDA, evaluative polarity from sentiment analysis, and contextual interpretation from representative posts. As a result, the driver-barrier classification is not only quantitatively grounded but also substantively meaningful for policy-oriented analysis.

3.5 Phase 5: Policy-Oriented Analysis

3.5.1 Topic salience and policy relevance

To translate public discourse into policy implications, each topic was analyzed according to its salience and evaluative orientation. Topic salience refers to the relative prominence of a topic in the corpus, while evaluative orientation refers to whether the topic reflects support or concern regarding electric vehicle adoption. Positive topics with high salience were interpreted as areas that should be reinforced through policy support, communication strategies, or ecosystem strengthening. Negative topics with high salience were interpreted as urgent policy issues because they represent barriers widely discussed by the public.

3.5.2 Policy priority index

To identify priority areas for policy intervention, a policy priority index was formulated by combining topic salience and the intensity of negative perception. The index is defined as:

$$PPI_j = \text{Norm}(F_j) \times \text{Norm}(N_j)$$

where, PPI_j denotes the policy priority index for topic j ; F_j represents the frequency or prevalence of topic j in the corpus; and N_j denotes the negative sentiment intensity associated with topic j . The function $\text{Norm}(\cdot)$ transforms each variable into a comparable scale ranging from 0 to 1.

The normalization function is defined as:

$$\text{Norm}(X_j) = \frac{X_j - \min(X)}{\max(X) - \min(X)}$$

where, X_j is the value of variable X for topic j ; while $\min(X)$ and $\max(X)$ denote the minimum and maximum values of that variable across all topics.

Accordingly, the normalized topic frequency and normalized negative sentiment intensity can be written as:

$$\text{Norm}(F_j) = \frac{F_j - \min(F)}{\max(F) - \min(F)}$$

$$\text{Norm}(N_j) = \frac{N_j - \min(N)}{\max(N) - \min(N)}$$

Thus, the policy priority index for topic j can be expressed explicitly as:

$$PPI_j = \left(\frac{F_j - \min(F)}{\max(F) - \min(F)} \right) \left(\frac{N_j - \min(N)}{\max(N) - \min(N)} \right)$$

Under this formulation, a topic obtains a high policy priority score when it appears frequently in public discourse and is simultaneously associated with strong negative sentiment. Therefore, topics with higher PPI_j values are interpreted as more urgent issues requiring policy attention, whereas topics with lower values are considered less critical. A higher policy priority index value indicates a more urgent policy issue because the topic is both prominent in public discourse and strongly associated with negative perceptions. Based on this logic, topics were grouped into three policy categories:

- (i) High-priority barriers, requiring immediate policy attention;
- (ii) Strategic drivers, requiring reinforcement and expansion;
- (iii) Emerging issues, requiring monitoring and gradual intervention.

This approach enables the study to move beyond descriptive sentiment mapping and toward evidence-based transport policy analysis.

3.5.3 Policy interpretation

The final stage interprets the identified drivers and barriers in relation to electric vehicle ecosystem development in Indonesia. The discussion links the topics to policy domains such as charging infrastructure, fiscal incentives, public awareness, battery ecosystem readiness, and regulatory consistency. In this way, the study transforms social media discourse into practical insights for transportation planners, policymakers, and electric vehicle ecosystem stakeholders.

4. Results and Discussion

This study investigates public perceptions of electric vehicle adoption in Indonesia by analyzing social media discourse related to electric mobility. The results are presented in four stages, namely data collection and dataset composition, sentiment classification, topic modeling, and policy-oriented interpretation. In general, the findings indicate that public discourse on electric vehicle adoption in Indonesia is dominated by neutral conversations, but still contains substantial negative concerns related to infrastructure, affordability, transition readiness, and policy implementation. At the same time, positive discourse highlights environmental benefits, technological attractiveness, energy efficiency, and the growing electric vehicle ecosystem. These results suggest that electric vehicle adoption in Indonesia is shaped by both encouraging drivers and persistent barriers, making a policy-oriented interpretation essential for translating public discourse into actionable transport policy recommendations.

4.1 Phase 1: Online Data Collection and Dataset Composition

A total of 6,844 cleaned social media posts related to electric vehicles were obtained for the period from 1 January to 31 December 2024. The dataset was collected using five keywords, namely electric cars, electric motorcycles, electric buses, electric vehicles, and electric vehicle batteries. Among these categories, the keyword electric vehicles produced the largest number of posts, amounting to 1,924 entries, followed by electric cars with 1,635 posts, electric motorcycles with 1,474 posts, electric vehicle batteries with 910 posts, and electric buses with 901 posts. This distribution indicates that electric vehicle discourse in Indonesia is not limited to private vehicle ownership, but also extends to two-wheeled mobility, public transport, and battery-related concerns. It also suggests that the discussion is spread across both user-oriented and ecosystem-oriented issues.

The composition of the dataset further shows that public discourse was concentrated most strongly on general electric vehicle terminology and private mobility. This is important because it reflects a public conversation that is still evolving conceptually: the term electric vehicle captures broad perceptions and policy discourse, whereas electric cars, electric motorcycles/bikes, and electric buses are more closely associated with practical adoption experiences. In contrast, the battery-related keyword appears in a smaller but still meaningful proportion of the corpus, indicating that battery issues remain a specialized but important topic within the electric vehicle ecosystem. From a transport policy perspective, this distribution suggests that electric vehicle adoption is discussed not only as a product issue but also as a system-level issue involving transport modes, infrastructure, and supporting technology.

This result supports the subsequent analysis because it shows that the dataset contains both adoption-level and ecosystem-level discourse. Therefore, the sentiment and topic-modeling results should be interpreted as a reflection of public evaluation of the broader electric vehicle transition rather than only individual vehicle purchase intention.

4.2 Phase 2: Sentiment Classification Results

Sentiment classification was conducted to identify whether electric vehicle-related discourse reflected supportive, neutral, or critical attitudes. The results show that social media conversations in 2024 were dominated by neutral sentiment, followed by negative and positive sentiment. This pattern indicates that public discussions were still largely observational and evaluative rather than fully polarized. Overall, the dataset contained 1,003 positive posts, 4,318 neutral posts, and 1,523 negative posts.

As presented in Table 2, neutral sentiment accounted for 63.09% of all posts, while negative sentiment reached 22.25% and positive sentiment accounted for 14.66%. This indicates that the public is still in a stage of observing, assessing, and discussing electric vehicle issues, rather than uniformly supporting or rejecting electric vehicle adoption. Nevertheless, the proportion of negative sentiment was considerably higher than positive sentiment, suggesting that public concerns remain substantial and should be addressed in transport policy design.

Table 2. Overall sentiment distribution of electric vehicle-related discourse

Sentiment	Number of Posts	Percentage (%)
Positive	1,003	14.66
Neutral	4,318	63.09
Negative	1,523	22.25
Total	6,844	100.00

A more detailed breakdown by keyword is presented in Figure 1 and Table 3. The figure shows that neutral discourse dominated all keyword groups. However, negative sentiment remained substantial for electric cars, electric motorcycles/bikes, and electric buses, indicating that practical concerns regarding electric vehicle adoption were still prominent in vehicle-oriented discussions.

Figure 1 and Table 3 show that electric vehicles generated the largest neutral volume, implying that broader electric vehicle discourse may still be driven by general information sharing, news, and public discussion rather than direct usage experience. At the same time, electric motorcycles recorded the largest negative volume relative to their category size, followed closely by electric cars. This suggests that two-wheeled and private electric vehicle adoption may be more strongly associated with practical issues such as affordability, usability, and infrastructure readiness. Meanwhile, electric vehicle batteries showed the lowest negative count, indicating that battery-related discourse was relatively less critical than vehicle-use discourse.

From a policy perspective, this sentiment pattern is meaningful. The dominance of neutrality suggests that the Indonesian public is still in an evaluation stage, where electric vehicles are seen as relevant but not yet fully normalized in everyday mobility. The persistent negative sentiment in vehicle-oriented keywords indicates that many users still associate electric vehicle adoption with unresolved practical barriers, including charging infrastructure, cost, and convenience. Therefore, the results imply that future policy interventions should focus

not only on increasing awareness, but also on reducing real-world adoption frictions experienced by potential users. The sentiment results also justify the need for topic modeling. The larger share of neutral posts indicates that descriptive information and news-sharing remain common in electric vehicle discourse, while the meaningful share of negative posts shows that unresolved barriers are still visible. Separating the corpus into positive and negative subsets therefore helps identify which themes function as adoption drivers and which themes function as adoption barriers.

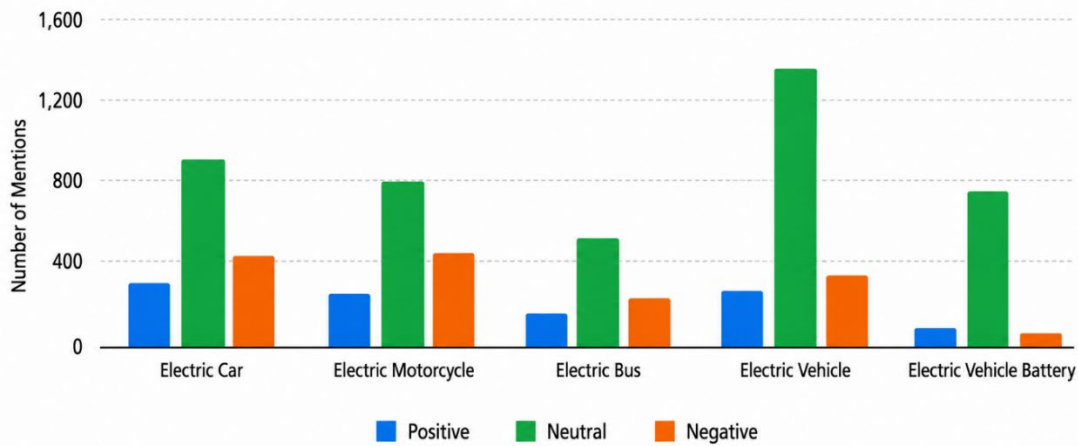


Figure 1. Sentiment distribution by electric vehicle keyword

Table 3. Sentiment distribution by keyword

Keyword	Positive	Neutral	Negative	Total
Electric car	296	906	433	1,635
Electric motorcycle	232	797	445	1,474
Electric bus	150	518	233	901
Electric vehicle	242	1,346	336	1,924
Electric vehicle battery	83	751	76	910
Total	1,003	4,318	1,523	6,844

4.3 Topic Modeling Results

Topic modeling was conducted to identify the dominant themes underlying positive and negative discourse on electric vehicle adoption in Indonesia. The results show that the positive corpus generated 10 optimum topics, while the negative corpus also generated 10 optimum topics. In general, the topic structure indicates that public perceptions of electric vehicle adoption in Indonesia are shaped by two major dimensions. On the one hand, positive discourse highlights environmental benefits, ecosystem development, accessibility, comfort, and efficiency. On the other hand, negative discourse reflects practical concerns related to user experience, transition barriers, policy criticism, skepticism toward innovation, and dissatisfaction with incentives.

4.3.1 Interpreted topics from the positive corpus

The positive-topic model produced ten interpretable themes that reflect the main drivers of electric vehicle adoption in Indonesia. These topics indicate that positive public discourse is largely associated with the image of electric vehicles as environmentally friendly, modern, accessible, and increasingly relevant to future mobility.

The positive-topic structure indicates that electric vehicle adoption in Indonesia is mainly encouraged by environmental narratives, infrastructure development, ecosystem progress, consumer preference, and perceived efficiency. These topics suggest that positive public opinion is driven not only by economic considerations, but also by symbolic and strategic meanings associated with clean, modern, and future-oriented mobility. Therefore, these themes can be interpreted as adoption drivers that should be reinforced through infrastructure visibility, public communication, and ecosystem development policies.

Table 4 and Table 5 should therefore be read together. Table 4 presents the detailed thematic structure of positive discourse, while Table 5 condenses these themes into policy-relevant adoption drivers. The dominance of environmental, infrastructure, ecosystem, preference, accessibility, comfort, and efficiency themes indicates that supportive public discourse is closely connected to the perceived usefulness of electric vehicles and the readiness

of supporting systems. These results provide the basis for reinforcing electric vehicle communication and infrastructure policies.

Table 4. Interpreted positive topics from the Latent Dirichlet Allocation (LDA) model

Positive Corpus	Optimum Topics = 10
Topic 1	Electric vehicles as environmentally friendly and the future of transportation
Topic 2	Public electric vehicle charging infrastructure and the future of battery-based electric vehicles in Indonesia
Topic 3	The development of environmentally friendly electric vehicles for the future
Topic 4	The development of the electric vehicle ecosystem in Indonesia
Topic 5	The role of electric bicycles and public electric vehicle charging infrastructure in creating more comfortable mobility
Topic 6	Consumer preferences toward battery-based electric vehicles in Indonesia
Topic 7	Accessibility of electric vehicles and electric bicycles in Indonesia
Topic 8	Battery-based motor vehicles as comfortable and environmentally friendly transport
Topic 9	Adoption of electric bicycles and the electric vehicle ecosystem in urban Indonesia
Topic 10	The increasingly economical use of electric vehicles in Indonesia

Table 5. Main positive topics interpreted as adoption drivers

Topic Code	Positive Topic	Policy Meaning
P1	Electric vehicles as environmentally friendly and future transportation	Strengthen environmental and future-mobility narratives
P2	Public electric vehicle charging infrastructure development	Expand visible infrastructure support
P3	Electric vehicle ecosystem development in Indonesia	Support ecosystem coordination and industrial readiness
P4	Consumer preference toward electric vehicles	Encourage public acceptance and market confidence
P5	Electric vehicle accessibility, comfort, and efficiency	Reinforce convenience and user experience

4.3.2 Negative topics as adoption barriers

The negative topics highlight the main barriers constraining electric vehicle adoption (Table 6). These include negative user perceptions and experiences, transition difficulties from conventional transport to electric vehicles, public criticism of government policy, skepticism toward electric vehicle innovation, and dissatisfaction with taxes and incentives.

Table 6. Main negative topics interpreted as adoption barriers

Topic Code	Negative Topic	Policy Meaning
N1	Negative user perceptions and experiences	Improve actual usage quality and service reliability
N2	Transition barriers from conventional transport to electric vehicles	Reduce practical conversion barriers
N3	Public criticism of electric vehicle-related government policies	Improve policy credibility and consistency
N4	Skepticism toward electric vehicle innovation	Strengthen trust-building and public education
N5	Dissatisfaction with taxes and incentives	Reform and clarify incentive mechanisms

These barrier topics indicate that electric vehicle adoption in Indonesia is still constrained by practical uncertainty and policy skepticism. Unlike the positive topics, which are largely future-oriented and aspirational, the negative topics reflect concrete adoption frictions. These include uncertainty about policy consistency, doubts about the usefulness of incentives, concern over transition readiness, and hesitation regarding actual user experience. Therefore, the main barrier to adoption does not appear to be rejection of the electric vehicle concept itself, but rather a lack of confidence in how the transition is being implemented and experienced in practice. Table 6 provides a more focused interpretation of negative discourse. The table shows that public resistance is not mainly directed at the idea of electric mobility itself. Instead, the barriers are related to implementation quality, transition difficulty, policy credibility, and the clarity of incentives. This finding is important because it suggests that policy intervention should prioritize practical adoption frictions rather than only promoting general awareness.

4.3.3 Driver–barrier interpretation

Taken together, the topic-modeling results show that electric vehicle adoption in Indonesia is driven by five main enabling narratives: environmental benefit, future mobility orientation, infrastructure development, comfort and accessibility, and perceived efficiency. At the same time, it is hindered by five major constraining narratives: negative user experience, transition difficulty, policy skepticism, incentive dissatisfaction, and consumer doubt. This pattern indicates that the Indonesian electric vehicle transition is not being contested at the level of basic desirability. Rather, it is being negotiated at the level of implementation, trust, and readiness. In other words, the public generally accepts the idea of electric mobility, but remains cautious about its current practicality and institutional support.

4.4 Policy-Oriented Discussion

From a policy-oriented perspective, the results indicate that the strongest adoption drivers should be reinforced, whereas the most persistent barriers should be directly addressed (Table 7). The positive-topic structure suggests that public support can be amplified through policies that emphasize electric vehicles as clean, efficient, and future-oriented mobility solutions. Infrastructure visibility, urban accessibility, comfort, and ecosystem development should therefore be treated as strategic strengths that need to be maintained and expanded. At the same time, the negative-topic structure indicates that the most urgent policy priorities lie in reducing adoption friction. The barrier themes suggest that practical experience, transition readiness, policy credibility, and incentive effectiveness remain critical. Therefore, the most immediate policy needs are likely to include improvement of charging accessibility, clearer and more predictable incentive mechanisms, better consumer communication regarding electric vehicle usage and benefits, and targeted support for user groups facing higher transition barriers.

Table 7. Proposed policy priority mapping

Priority Level	Main Issue	Suggested Intervention
High	Transition barriers and negative user experience	Improve charging access, after-sales support, and usability
High	Incentive dissatisfaction and policy skepticism	Clarify subsidy mechanisms and strengthen policy consistency
Medium	Infrastructure visibility and accessibility	Expand public electric vehicle charging infrastructure deployment and information transparency
Medium	Electric vehicle affordability concerns	Improve financing access and targeted support
Strategic strength	Environmental and future-mobility narratives	Reinforce positive communication and public campaigns
Strategic strength	Electric vehicle ecosystem development	Strengthen coordination among government, industry, and infrastructure providers

Based on the combined sentiment and topic interpretation, the most urgent policy issues are those related to transition barriers, negative user experience, and dissatisfaction with policy instruments. By contrast, environmental narratives, infrastructure progress, and ecosystem development can be treated as strategic strengths that should be reinforced. This supports a dual policy approach: corrective intervention for barriers and strategic reinforcement for drivers.

Table 7 operationalizes the driver-barrier interpretation into policy priorities. High-priority issues are assigned to topics that combine negative public perception with practical relevance for adoption, such as user experience, transition barriers, and incentive dissatisfaction. Medium-priority issues represent enabling conditions that still require improvement, while strategic strengths represent positive narratives that should be maintained through communication, infrastructure visibility, and ecosystem coordination.

4.5 Comparison with Recent Studies

The findings are broadly consistent with recent studies on electric vehicle adoption in Indonesia. Hakam & Jumayla (2024) and Lazuardy et al. (2024) emphasized that infrastructure readiness, affordability, regulatory consistency, and technological concerns remain major barriers to electric vehicle diffusion. Similar issues appear in the negative topics of this study, particularly transition barriers, policy skepticism, incentive dissatisfaction, and concerns about user experience. This similarity indicates that online public discourse reflects many of the same structural barriers identified in survey-based and policy-oriented studies. The results also align with recent social-media-based electric vehicle studies. Nugroho & Widiyanto (2024) showed that Twitter discourse in Indonesia contains important adoption-related aspects, while Qian & Gkritza (2024) and Wang et al. (2024) demonstrated that public perception of electric vehicles and charging infrastructure varies across regions and is influenced by infrastructure availability. Ashby et al. (2025) similarly highlighted charger reliability, accessibility, and charging

inequality as central public concerns. In line with these studies, the present findings show that infrastructure visibility and user experience are central elements in both positive and negative electric vehicle discourse.

However, this study differs from previous studies in its policy-oriented integration of sentiment analysis and topic modeling. Survey-based studies such as Astuti & Susanto (2024) and Basmantra et al. (2025) mainly explain behavioral intention and acceptance determinants, while many social media studies focus on sentiment or topic identification separately. This study extends those approaches by separating positive and negative corpora, interpreting them as adoption drivers and barriers, and translating the results into a policy priority mapping for Indonesia. Therefore, the contribution of this study lies not only in identifying what people discuss online but also in showing how online discourse can support evidence-based electric vehicle policy formulation.

4.6 Limitations of Social Media Data

This study also has limitations associated with the use of social media data. First, X users do not necessarily represent the whole Indonesian population. The platform may overrepresent digitally active, urban, younger, or politically engaged users, while underrepresenting people with limited internet access or low social media activity. Consequently, the findings should be interpreted as evidence of online public discourse rather than as a statistically representative measure of national public opinion. Second, social media datasets can be affected by inactive users, automated accounts, promotional posts, repeated content, and event-driven spikes in discussion. This study reduced these risks by removing duplicates, filtering non-informative posts, and excluding spam-like promotional content. However, some residual bias may remain because account-level demographic information and automated-account detection were not fully available. These limitations suggest that social media mining is best used as a complementary decision-support tool alongside surveys, interviews, market data, and stakeholder consultation.

5. Conclusion

This study investigated the drivers and barriers of electric vehicle adoption in Indonesia through a policy-oriented social media mining approach. By analyzing 6,844 public posts collected from X, this study demonstrated that social media discourse can provide valuable empirical evidence for understanding public perceptions of electric mobility in an emerging market context. The integration of IndoRoBERTa-based sentiment analysis and LDA topic modeling enabled the study to identify not only the overall sentiment orientation of electric vehicle-related discussions but also the specific themes that shape public support and concern toward electric vehicle adoption. The findings show that electric vehicle discourse in Indonesia is still dominated by neutral sentiment, indicating that the public is largely in an evaluation stage rather than a fully supportive or resistant stage. However, the proportion of negative sentiment was higher than positive sentiment, suggesting that unresolved concerns remain important barriers to adoption. Topic modeling further revealed that positive discourse is mainly driven by environmental benefits, future-mobility narratives, charging infrastructure development, electric vehicle ecosystem growth, accessibility, comfort, and perceived operational efficiency. These themes indicate that electric vehicles are viewed positively when they are associated with clean mobility, technological progress, and long-term transportation transformation.

At the same time, the negative topics highlight several critical adoption barriers. These include negative user experiences, transition difficulties from conventional vehicles to electric vehicles, skepticism toward electric vehicle-related policies, doubts regarding technological innovation, and dissatisfaction with taxes and incentive mechanisms. These findings suggest that the main challenge for electric vehicle adoption in Indonesia is not rejection of the electric vehicle concept itself, but limited confidence in implementation readiness, policy consistency, infrastructure accessibility, and practical user experience. From a policy perspective, the results support a dual intervention strategy. First, policymakers should reinforce existing adoption drivers by strengthening environmental communication, expanding visible charging infrastructure, improving public awareness, and supporting electric vehicle ecosystem coordination among government, industry, electricity providers, and infrastructure operators. Second, policymakers should directly address adoption barriers by improving charging access, after-sales support, user reliability, subsidy transparency, financing accessibility, and policy credibility. In this regard, social media mining can function as a decision-support tool for identifying public concerns and translating them into actionable policy priorities.

This study contributes to the literature by extending electric vehicle adoption research beyond conventional survey-based approaches. It provides large-scale evidence of public perceptions in Indonesia and demonstrates how sentiment analysis and topic modeling can be combined to identify adoption drivers, barriers, and policy priorities. Practically, the findings offer useful insights for transportation planners, policymakers, electric vehicle manufacturers, charging infrastructure providers, and other ecosystem stakeholders seeking to accelerate the electric mobility transition in Indonesia. Future research can extend this study in several directions. First, additional platforms such as YouTube, Facebook, Instagram, TikTok, online forums, and online news comments

can be analyzed to capture broader public discourse. Second, cross-country comparisons can be conducted to examine whether electric vehicle adoption drivers and barriers differ across emerging markets, especially in Southeast Asia. Third, a longer longitudinal observation period can be used to evaluate how public perception changes before and after major policy interventions, infrastructure expansion, or electric vehicle market events. Fourth, future studies can combine social media mining with surveys, interviews, or stakeholder validation to improve representativeness and strengthen the policy relevance of the findings.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

References

- Acheampong, R. A. & Cugurullo, F. (2019). Capturing the behavioural determinants behind the adoption of autonomous vehicles: Conceptual frameworks and measurement models to predict public transport, sharing and ownership trends of self-driving cars. *Transp. Res. F: Traffic Psychol. Behav.*, 62, 349–375. <https://doi.org/10.1016/j.trf.2019.01.009>.
- Alshahapy, H., Bozeman, J. F., Carley, S., Nock, D., & Matisoff, D. (2025). Modern challenges facing electric vehicle adoption: A review of barriers to adoption, supply chain challenges, and equity. *Environ. Res. Lett.*, 20(2), 023002. <https://doi.org/10.1088/1748-9326/ada6de>.
- Anugerah, A. R., Muttaqin, P. S., & Purnama, D. A. (2021). Effect of large-scale social restriction (PSBB) during COVID-19 on outdoor air quality: Evidence from five cities in DKI Jakarta Province, Indonesia. *Environ. Res.*, 197, 111164. <https://doi.org/10.1016/j.envres.2021.111164>.
- Asamer, J., Graser, A., Heilmann, B., & Ruthmair, M. (2016). Sensitivity analysis for energy demand estimation of electric vehicles. *Transp. Res. D: Transp. Environ.*, 46, 182–199. <https://doi.org/10.1016/j.trd.2016.03.017>.
- Ashby, C., Weir, D., & Fussey, P. (2025). Understanding public views on electric vehicle charging: A thematic analysis. *Transp. Res. Interdiscip. Perspect.*, 29, 101325. <https://doi.org/10.1016/j.trip.2025.101325>.
- Astuti, R. D. & Susanto, A. A. (2024). Challenges of electric vehicle adoption in Indonesia: Revealing the hidden factors affecting purchase intention. *J. Siasat Bisnis*, 149–171. <https://doi.org/10.20885/jsb.vol28.iss2.art2>.
- Bachtiar, W. F., Masruroh, N. A., Asih, A. M. S., & Sari, D. P. (2024). Halal food sustainable traceability framework for the meat processing industry. *J. Islam. Mark.*, 15(11), 2759–2784. <https://doi.org/10.1108/jima-12-2023-0412>.
- Balla, S. N., Pani, A., Sahu, P. K., & González-Feliu, J. (2023). Examining shifts in public discourse on electric mobility adoption through Twitter data. *Transp. Res. D: Transp. Environ.*, 121, 103843. <https://doi.org/10.1016/j.trd.2023.103843>.
- Bansal, A., Pranav, K., & Bhopale, M. (2024). A comprehensive life cycle assessment of an LNG carrier from cradle to grave. In *Abu Dhabi International Petroleum Exhibition and Conference* (p. D041S137R004). SPE.
- Basmantra, I. N., Santiarsa, N. K. S., Widodo, R. D., & Mimaki, C. A. (2025). How the adoption of EVs in developing countries can be effective: Indonesia's case. *World Electr. Veh. J.*, 16(8), 428. <https://doi.org/10.3390/wevj16080428>.
- Choi, J., Oh, S., Yoon, J., Lee, J., & Coh, B. (2020). Identification of time-evolving product opportunities via social media mining. *Technol. Forecast. Soc. Change*, 156, 120045. <https://doi.org/10.1016/j.techfore.2020.120045>.
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Q.*, 13(3), 319–340. <https://doi.org/10.2307/249008>.
- Degirmenci, K. & Breitner, M. H. (2017). Consumer purchase intentions for electric vehicles: Is green more important than price and range? *Transp. Res. D: Transp. Environ.*, 51, 250–260. <https://doi.org/10.1016/j.trd.2017.01.001>.
- Hakam, D. F. & Jumayla, S. (2024). Electric vehicle adoption in Indonesia: Lesson learned from developed and developing countries. *Sustain. Futures*, 8, 100348. <https://doi.org/10.1016/j.sfr.2024.100348>.
- Hardman, S., Chandan, A., Tal, G., & Turrentine, T. (2017). The effectiveness of financial purchase incentives for battery electric vehicles—A review of the evidence. *Renew. Sust. Energ. Rev.*, 80, 1100–1111. <https://doi.org/10.1016/j.rser.2017.05.255>.
- IEA. (2024). *Global EV outlook 2024*. IEA, Paris. <https://www.iea.org/reports/global-ev-outlook-2024>

- Jeong, B., Yoon, J., & Lee, J. (2019). Social media mining for product planning: A product opportunity mining approach based on topic modeling and sentiment analysis. *Int. J. Inf. Manag.*, 48, 280–290. <https://doi.org/10.1016/j.ijinfomgt.2017.09.009>.
- Lazuardy, A., Nurcahyo, R., Kristiningrum, E., Ma'aram, A., Farizal, Aqmarina, S. N., & Rajabi, M. F. (2024). Technological, environmental, economic, and regulation barriers to electric vehicle adoption: Evidence from Indonesia. *World Electr. Veh. J.*, 15(9), 422. <https://doi.org/10.3390/wevj15090422>.
- Li, W., Long, R., Chen, H., & Geng, J. (2017). A review of factors influencing consumer intentions to adopt battery electric vehicles. *Renew. Sust. Energ. Rev.*, 78, 318–328. <https://doi.org/10.1016/j.rser.2017.04.076>.
- Nugroho, S. A. & Widiyanto, S. (2024). Exploring electric vehicle adoption in Indonesia using zero-shot aspect-based sentiment analysis. *Sustain. Oper. Comput.*, 5, 191–205. <https://doi.org/10.1016/j.susoc.2024.08.002>.
- Purnama, D. A., Lumi, D. P., Febriani, A., Utama T, A. R., Salindri, S. D., Anugerah, A. R., & Aliafari, N. (2025). Data-driven public policy for electric vehicles (EV) through open innovation and dynamic consumer preferences: A time-series social media analysis using integrated IPA-product improvability model. *J. Open Innov.: Technol. Mark. Complex.*, 11(3), 100583. <https://doi.org/10.1016/j.joitmc.2025.100583>.
- Qian, X. & Gkritza, K. (2024). Spatial and temporal variance in public perception of electric vehicles: A comparative analysis of adoption pioneers and laggards using Twitter data. *Transp. Policy*, 149, 150–162. <https://doi.org/10.1016/j.tranpol.2024.02.011>.
- Rezvani, Z., Jansson, J., & Bodin, J. (2015). Advances in consumer electric vehicle adoption research: A review and research agenda. *Transp. Res. D: Transp. Environ.*, 34, 122–136. <https://doi.org/10.1016/j.trd.2014.10.010>.
- Sierzchula, W., Bakker, S., Maat, K., & van Wee, B. (2014). The influence of financial incentives and other socio-economic factors on electric vehicle adoption. *Energy Policy*, 68, 183–194. <https://doi.org/10.1016/j.enpol.2014.01.043>.
- Stieglitz, S., Mirbabaie, M., Ross, B., & Neuberger, C. (2018). Social media analytics—Challenges in topic discovery, data collection, and data preparation. *Int. J. Inf. Manag.*, 39, 156–168. <https://doi.org/10.1016/j.ijinfomgt.2017.12.002>.
- Subagyo, Purnama, D. A., Masrurroh, N. A., & Pratama, R. R. (2024). Modeling dynamic consumer preferences in product attributes for social media-based product improvement planning. *Malays. J. Consum. Fam. Econ.*, 32(1), 104–140. <https://doi.org/10.60016/majcafe.v32.05>.
- Taamneh, M. M. & Makahleh, H. Y. (2025). The prospects of adopting electric vehicles in urban contexts: A systematic review of literature. *Transp. Res. Interdiscip. Perspect.*, 31, 101420. <https://doi.org/10.1016/j.trip.2025.101420>.
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view 1. *MIS Q.*, 27(3), 425–478. <https://doi.org/10.2307/30036540>.
- Wang, J., Lai, J., & Lin, Y. (2023). Social media analytics for mining customer complaints to explore product opportunities. *Comput. Ind. Eng.*, 178, 109104. <https://doi.org/10.1016/j.cie.2023.109104>.
- Wang, M., Zhao, L., & Cochran, A. L. (2024). Sentiments of rural U.S. communities on electric vehicles and infrastructure: Insights from Twitter data. *Sustainability*, 16(11), 4871. <https://doi.org/10.3390/su16114871>.
- Zhao, Y., Zhang, C., Zhang, Y., Wang, Z., & Li, J. (2020). A review of data mining technologies in building energy systems: Load prediction, pattern identification, fault detection and diagnosis. *Energy Built Environ.*, 1(2), 149–164. <https://doi.org/10.1016/j.enbenv.2019.11.003>.